



Machine Learning & FinTech

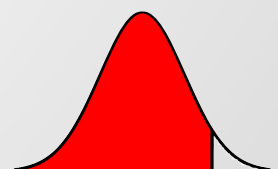
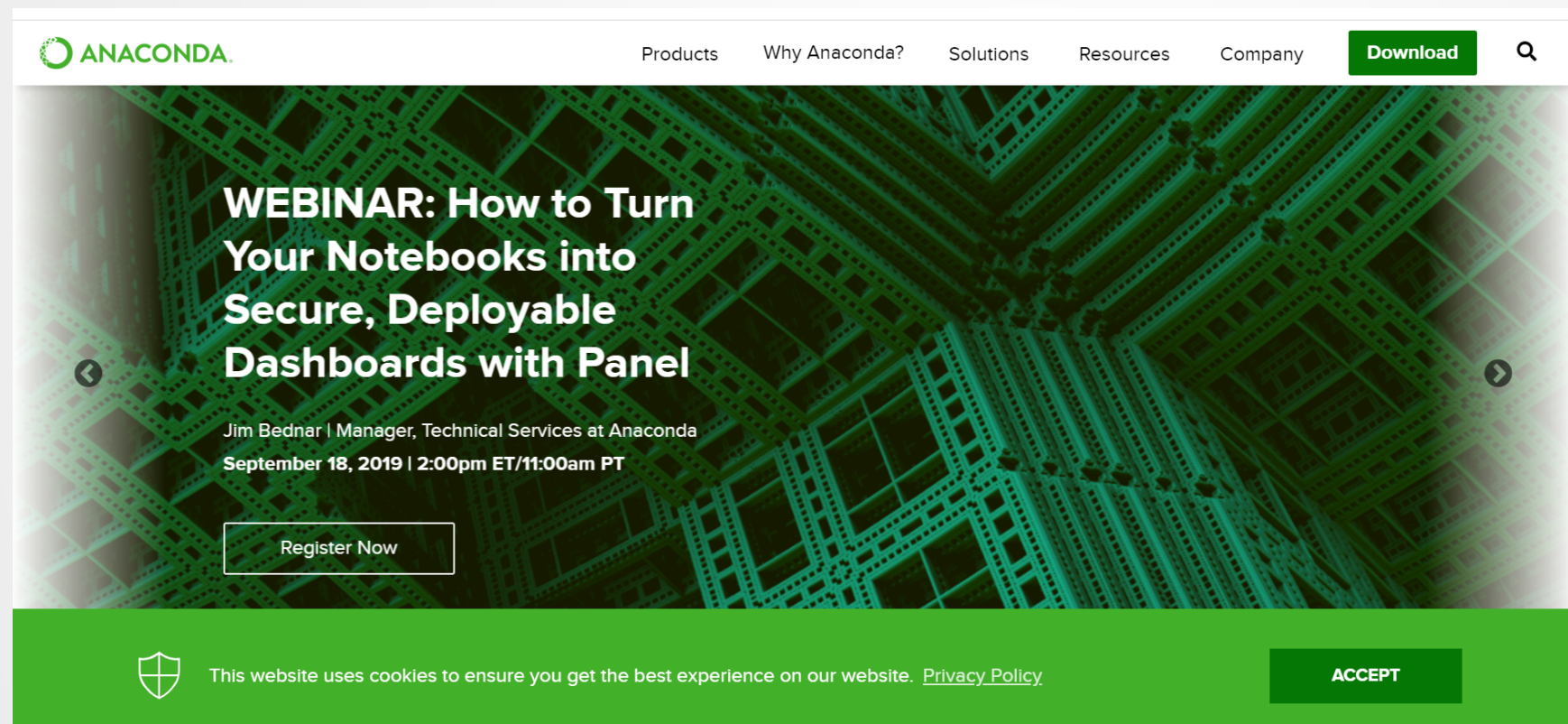
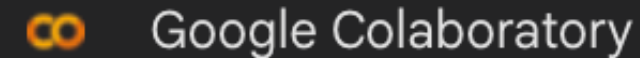
Python and EDA

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<https://hackmd.io/@hwteng/HyKOPoA6d>

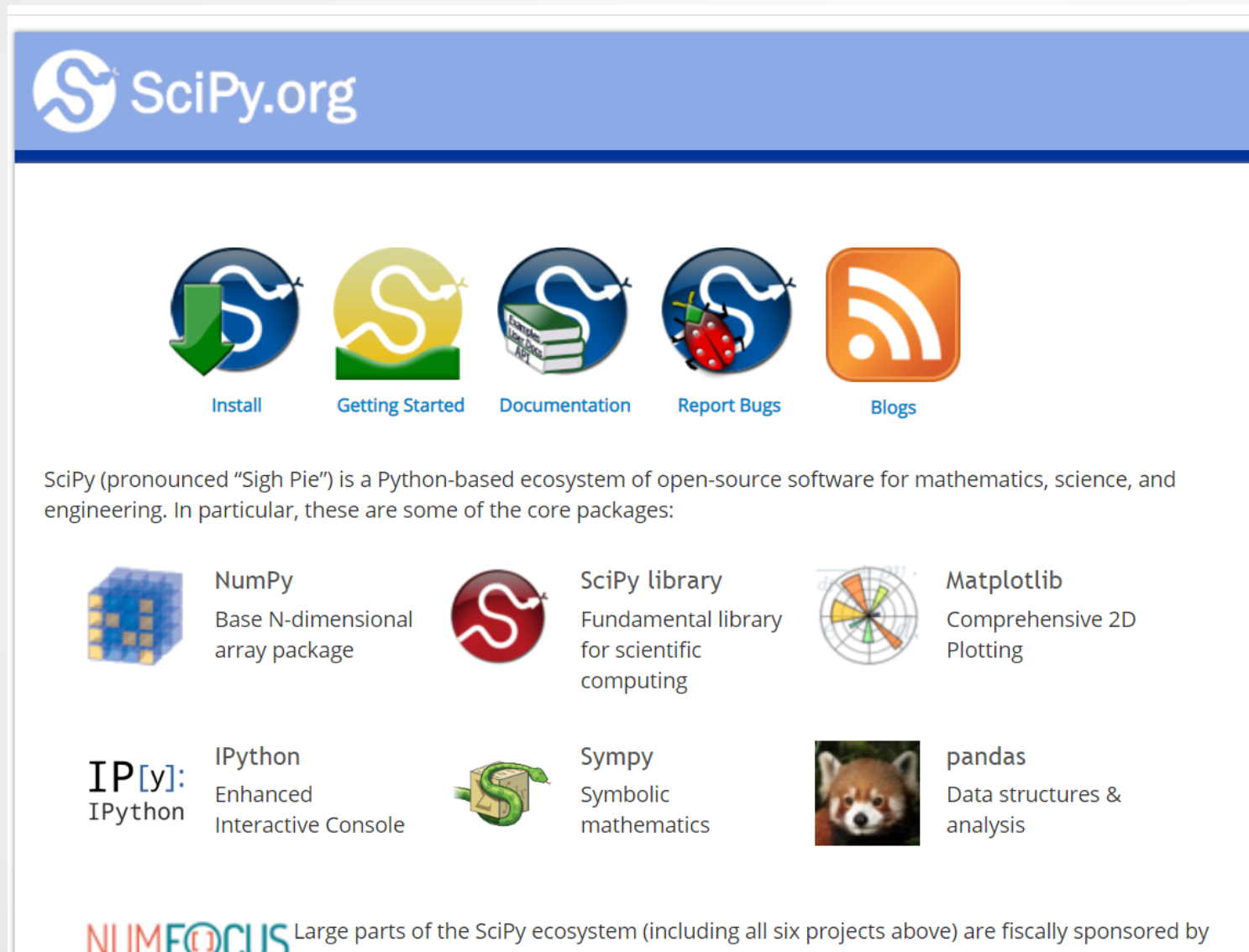
Motivation

- Anaconda, Visual Studio Code, or others
 - .py (Spyder, similar to Matlab)
 - .ipynb (jupyter, used for course demonstrations; Colab in google)



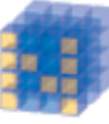
SciPy

- SciPy (pronounced “Sigh Pie”) is a Python-based ecosystem of open-source software for mathematics, science, and engineering.



The screenshot shows the SciPy.org website. At the top is the SciPy.org logo. Below it are five navigation icons: Install (a blue circle with a green arrow), Getting Started (a yellow circle with a white 'S'), Documentation (a blue circle with a white 'S' and a book icon), Report Bugs (a blue circle with a white 'S' and a red bug icon), and Blogs (an orange RSS icon). Below these icons is a paragraph: "SciPy (pronounced “Sigh Pie”) is a Python-based ecosystem of open-source software for mathematics, science, and engineering. In particular, these are some of the core packages:". Below this paragraph are six core packages arranged in two rows. Each package has an icon, a name, and a description. The first row includes NumPy (a blue cube icon, "Base N-dimensional array package"), SciPy library (a red circle with a white 'S' icon, "Fundamental library for scientific computing"), and Matplotlib (a circular plot icon, "Comprehensive 2D Plotting"). The second row includes IPython (a blue 'IP[y]:' icon, "Enhanced Interactive Console"), SymPy (a green cube icon, "Symbolic mathematics"), and pandas (a red panda face icon, "Data structures & analysis"). At the bottom of the screenshot, there is a logo for NUMFOCUS and a line of text: "Large parts of the SciPy ecosystem (including all six projects above) are fiscally sponsored by".

SciPy (pronounced “Sigh Pie”) is a Python-based ecosystem of open-source software for mathematics, science, and engineering. In particular, these are some of the core packages:

	NumPy Base N-dimensional array package		SciPy library Fundamental library for scientific computing		Matplotlib Comprehensive 2D Plotting
	IPython Enhanced Interactive Console		Sympy Symbolic mathematics		pandas Data structures & analysis

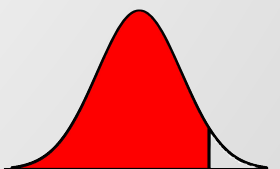
NUMFOCUS Large parts of the SciPy ecosystem (including all six projects above) are fiscally sponsored by

- Reading: Scipy Lecture Notes



NymPy

- ▣ NumPy is the fundamental package for scientific computing with Python. It contains among other things:
 - ▣ a powerful N-dimensional array object
 - ▣ sophisticated (broadcasting) functions
 - ▣ tools for integrating C/C++ and Fortran code
 - ▣ useful linear algebra, Fourier transform, and random number capabilities
- ▣ Reading: [Numpy Quick Start Tutorials](#)



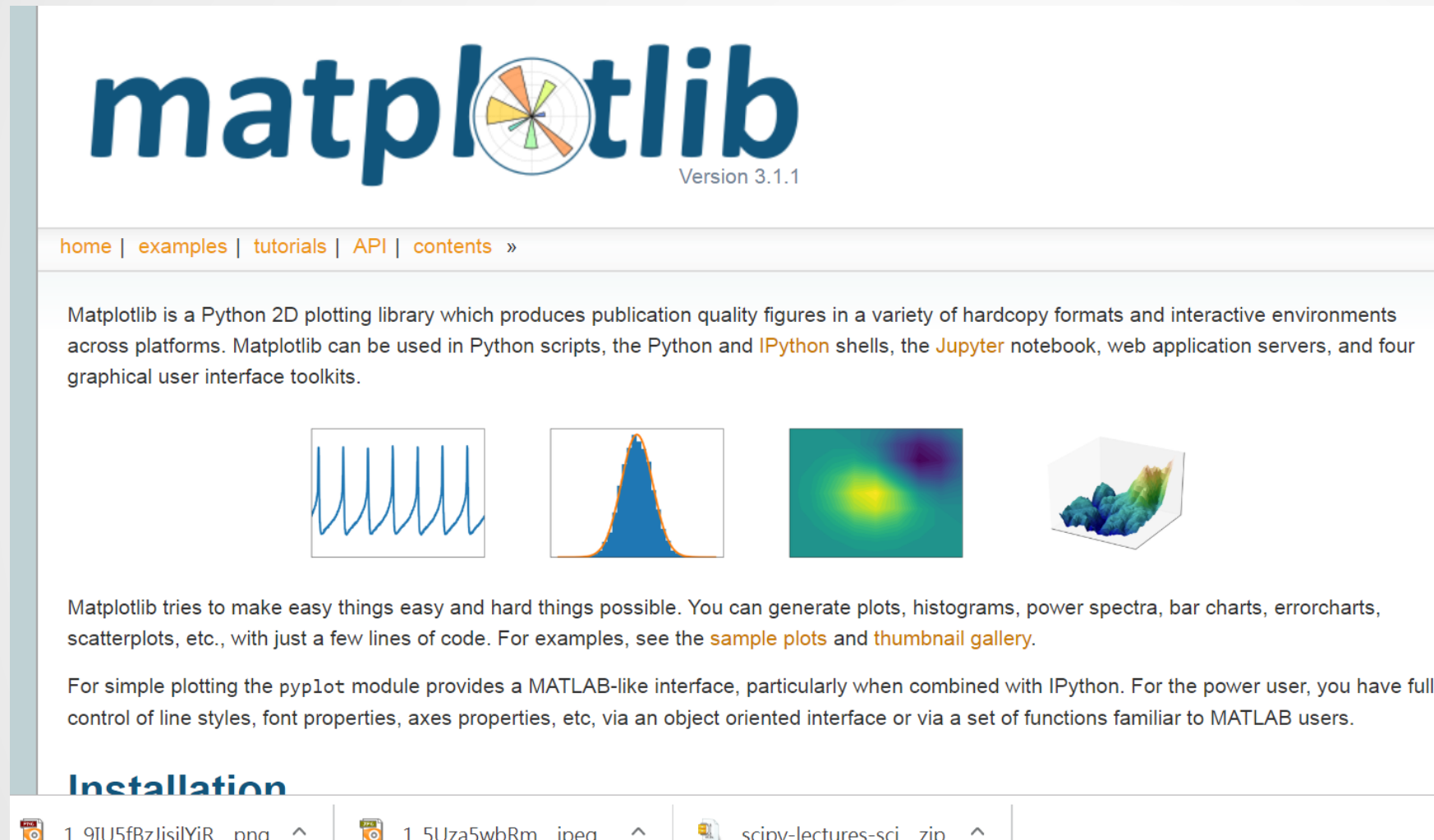
Pandas

- ▣ Pandas stands for “Python Data Analysis Library”.
According to the Wikipedia page on Pandas, “the name is derived from the term “panel data”, an econometrics term for multidimensional structured data sets.”
- ▣ Pandas is for data munging
- ▣ It provides Series and DataFrame data structures



matplotlib

- Matplotlib is a Python 2D plotting library.



The screenshot shows the Matplotlib website homepage. At the top is the Matplotlib logo, which consists of the word "matplotlib" in a blue serif font, with a circular icon containing a stylized plot of a sine wave and a scatter plot. Below the logo is the text "Version 3.1.1". A navigation bar contains links: "home", "examples", "tutorials", "API", and "contents", followed by a right-pointing arrow. The main content area has a paragraph describing Matplotlib as a Python 2D plotting library that produces publication quality figures in various formats and environments. Below this paragraph are four small images: a line plot of a sine wave, a histogram with a normal distribution curve overlaid, a 2D heatmap, and a 3D surface plot. Another paragraph follows, stating that Matplotlib tries to make easy things easy and hard things possible, and lists various plot types it can generate. A final paragraph mentions the pyplot module for a MATLAB-like interface. The word "Installation" is visible at the bottom of the main content area. At the very bottom of the screenshot, there is a file manager interface showing three files: "1 9IU5fBzJisiYiR....png", "1 5Uza5wbRm....ipea", and "scipy-lectures-sci....zip".

matplotlib
Version 3.1.1

[home](#) | [examples](#) | [tutorials](#) | [API](#) | [contents](#) »

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and [IPython](#) shells, the [Jupyter](#) notebook, web application servers, and four graphical user interface toolkits.



Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, errorcharts, scatterplots, etc., with just a few lines of code. For examples, see the [sample plots](#) and [thumbnail gallery](#).

For simple plotting the `pyplot` module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

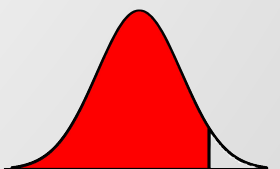
Installation

1 9IU5fBzJisiYiR....png ^ 1 5Uza5wbRm....ipea ^ scipy-lectures-sci....zip ^



Reading: Python basics and python cheat sheets

- ▣ #python_1_basics.ipynb
- ▣ #python_2_numPy.ipynb
- ▣ #python_3_pandas.ipynb



Exploratory Data Analysis (EDA)

- EDA was promoted by John Tukey (FFT and box plot) to encourage statisticians to **explore the data**, and possibly **formulate hypotheses** that could lead to new data collection and experiments.



https://en.wikipedia.org/wiki/John_Tukey



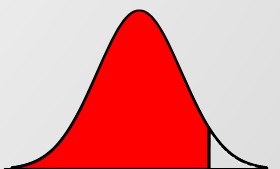
A nutshell

- ▣ The EDA tools allow researchers to obtain a general impression of their data.
 - ▣ Recall Chapters 1 to 3 in Statistics 101
 - ▣ **Simple numerical presentation:** mean, var, std, max, min, quantiles.
 - ▣ **Simple graphical presentation:** histogram, boxplots, scatterplots.
- ▣ What's more?
 - ▣ **Sophisticated numerical presentation:** statistical tests (ADF test, LM test)
 - ▣ **Sophisticated Graphical presentation:** dimensional reduction (PCA), clustering, etc.



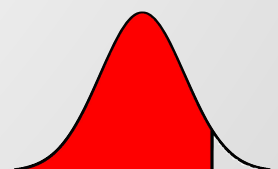
The objectives of EDA

- ▣ The objectives
 - ▶ Enable unexpected discoveries in the data
 - ▶ Suggest hypotheses about the causes of observed phenomena
 - ▶ Assess assumptions on which statistical inference will be based
 - ▶ Support the selection of appropriate statistical tools and techniques
 - ▶ Provide a basis for further data collection through surveys or experiments
- ▣ Closely related to: data mining, data exploration, data visualization, unsupervised learning!



Example

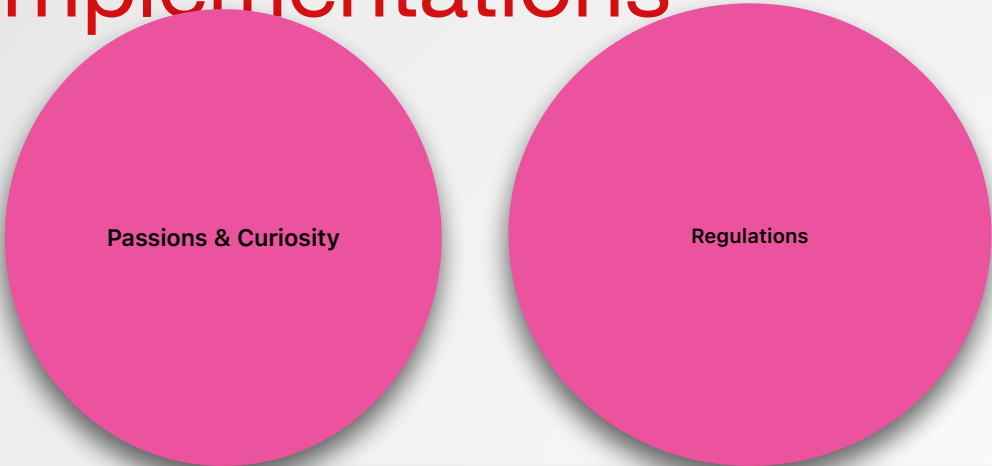
- ▣ This research aimed at the case of customer's default payments in Taiwan and compares the predictive accuracy of probability of default among six data mining methods.
- ▣ Yeh, I. C., & Lien, C. H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473-2480. <https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>.
- ▣ **#EDA_credit_default.ipynb**



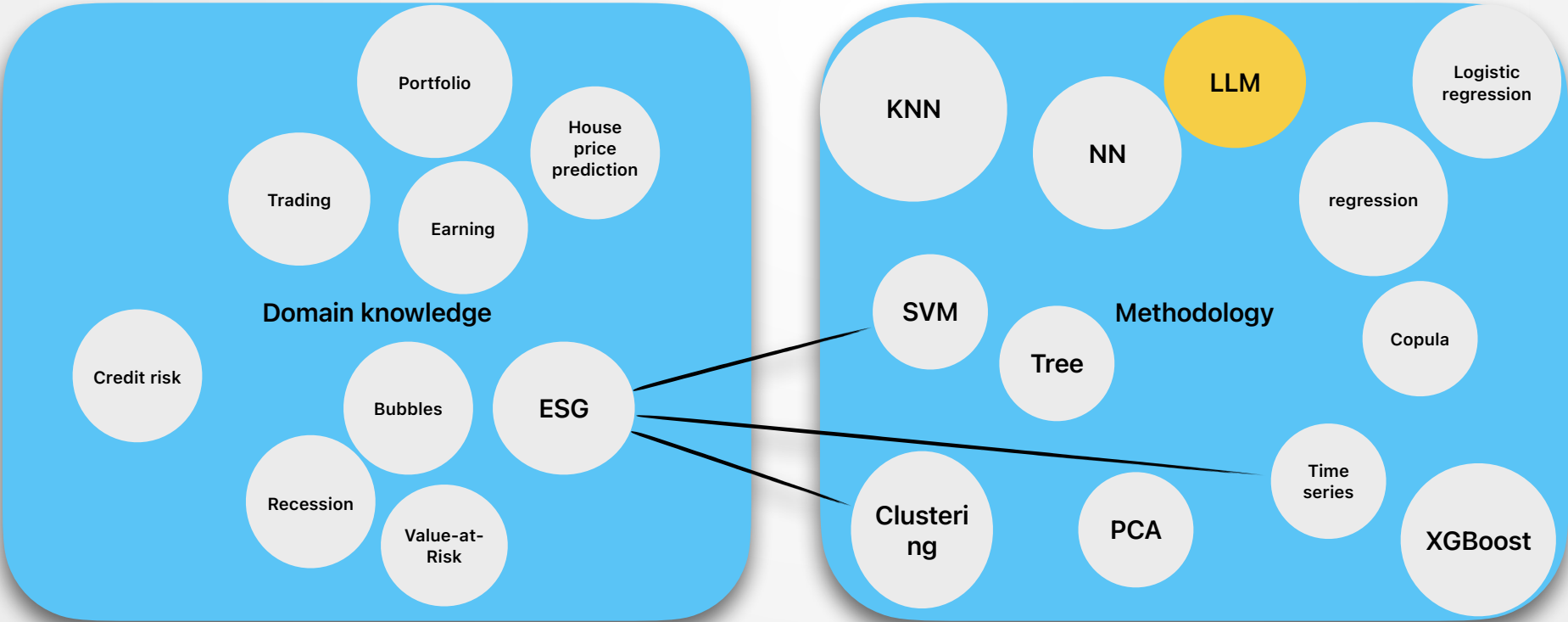
Policies & regulations



Implementations



Implementation



Source of data

- Kaggle
 - IEEE-CIS Fraud Detection
 - House Prices: Advanced Regression Techniques
- UCI Machine Learning Repository
 - Bank Marketing Dataset
 - Default of credit card clients dataset

NYCU paid dataset:

- **TEJ**
- **WRDS**



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